

Mini ERP + CRM Operations Portal

Comprehensive Project Explanation, Technical Architecture & Operational Guide

Executive Summary: The Mini ERP + CRM Operations Portal is a full-stack, enterprise-grade Operations Management Platform designed specifically for B2B wholesale and distribution companies. It addresses critical operational challenges in inventory control, customer relationship pipelines, order fulfillment, and role-based accountability.

1. Business Problem & Solution Objectives

Wholesale distribution businesses handle large-volume orders across multiple internal stakeholders: Sales Representatives, Warehouse Supervisors, and Accounts Officers. In traditional systems, two major issues repeatedly disrupt operations:

- **Phantom Stock & Uncontrolled Deductions:** When sales quotes or draft orders immediately deduct inventory, actual available stock becomes inaccurate, causing valid customer orders to be rejected.
- **Over-allocation & Negative Stock:** Confirming orders without strict real-time stock checks results in negative inventory, unfulfilled deliveries, and shipping delays.
- **Lack of CRM Audit Trail:** B2B sales cycles require scheduled follow-ups, interaction logs, and status transitions (LEAD → ACTIVE) which are easily lost without integrated CRM history.

System Core Objectives:

1. Enforce **atomic transaction-based inventory deduction** only when a Sales Challan is explicitly CONFIRMED.
2. Provide complete **Role-Based Access Control (RBAC)** for ADMIN, SALES, WAREHOUSE, and ACCOUNTS roles.
3. Store **historical product snapshots** in delivery line items to preserve historical price and description records.

2. Technology Stack & Architecture

Layer	Technologies Used	Purpose & Key Libraries
Frontend	React 18, TypeScript, Vite	Responsive SPA UI, React Router v6, Axios, Lucide Icons
Backend API	Node.js, Express.js, TypeScript	Modular REST APIs, Controller-Service architecture
Security & Auth	JWT, bcryptjs, Helmet, CORS	Stateless auth, hashed passwords, RBAC middleware
Database & ORM	PostgreSQL, Prisma ORM	Relational data modeling, schema migrations, \$transaction
Deployment	Render / Vercel / Neon	Production deployment configuration, Docker compose

System Flow & Control Architecture:

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React Frontend (TypeScript + Vite)
  ■■■ Axios Interceptor (Attaches Authorization: Bearer <JWT>)
    ■■■ Express REST API Server
      ■■■ authMiddleware (JWT Verification)
      ■■■ roleMiddleware (Role Permission Check)
      ■■■ Controllers & Services (Business Logic Execution)
      ■■■ Prisma ORM ■■■(Atomic $transaction)■■■> PostgreSQL Database
```

3. Database Schema & Data Model Design

The relational database schema is modeled in Prisma ORM (mapped to PostgreSQL). Key entity models and relations include:

Model Name	Key Attributes	Relationships & Constraints
User	id, name, email, password, role	Email unique. Roles: ADMIN, SALES, WAREHOUSE, ACCOUNTS.
Customer	id, name, mobile, businessName, customerType, status, followUpDate	Types: RETAIL, WHOLESALE, DISTRIBUTOR. Status: LEAD, ACTIVE, INACTIVE. Has many followUps & challans.
CustomerFollowUp	id, customerId, note, followUpDate, createdBy, createdAt	Belongs to Customer. Tracks sales CRM notes and next dates.
Product	id, name, sku, category, unitPrice, currentStock, minimumStock, location	SKU unique. Low stock rule: currentStock <= minimumStock.
StockMovement	id, productId, quantity, movementType, reason, createdBy, createdAt	MovementType: IN, OUT. Direct audit log of stock changes.
Challan	id, challanNumber, customerId, totalQuantity, status, createdBy	ChallanNumber unique (CH-00001). Status: DRAFT, CONFIRMED, CANCELLED.
ChallanItem	id, challanId, productId, productNameSnapshot, skuSnapshot, unitPriceSnapshot, quantity	Belongs to Challan & Product. Contains product snapshot fields.

4. Critical Challan Stock Transaction Logic

The core innovation of the portal is the **Transactional Challan Confirmation Engine**. It enforces zero stock leakage through an isolated database transaction.

Step 1: DRAFT Creation: When a sales representative creates a Challan, status is set to DRAFT. Line item snapshots (name, SKU, price) are stored into ChallanItem. **No stock is deducted at this stage.**

Step 2: Confirmation Trigger: When authorized staff invokes POST /api/challans/:id/confirm, the service opens a Prisma \$transaction block.

Step 3: Pre-Check Stock Availability: The transaction fetches all line items and compares product.currentStock against item.quantity for every product in the challan.

Step 4: Atomic Rejection Rule: If **ANY** product has insufficient stock, the transaction immediately throws an error and rolls back completely: Insufficient stock for [Name]. Available: X, Requested: Y (HTTP 400). Zero partial deductions occur.

Step 5: Atomic Deduction & Audit Log: If all items pass verification, the transaction deducts stock for every product, creates an OUT StockMovement audit record for each product line, updates Challan status to CONFIRMED, and commits.

5. Role-Based Access Control (RBAC) Matrix

Module / Endpoint	ADMIN	SALES	WAREHOUSE	ACCOUNTS
Authentication & Profile	Full Access	Full Access	Full Access	Full Access
Executive Dashboard Stats	Full Access	Full Access	Full Access	Full Access
Customer CRM List & Detail	Full Access	Full Access	View Only	Full Access
Delete Customer Record	ALLOWED	FORBIDDEN (403)	FORBIDDEN (403)	FORBIDDEN (403)
Add CRM Follow-up Note	ALLOWED	ALLOWED	FORBIDDEN (403)	ALLOWED
Product Inventory Catalog	Full Access	View Only	Full Access (CRUD)	View Only
Record Direct Stock IN/OUT	ALLOWED	FORBIDDEN (403)	ALLOWED	FORBIDDEN (403)
Create / Edit Draft Challans	ALLOWED	ALLOWED	FORBIDDEN (403)	FORBIDDEN (403)
Confirm Sales Delivery Challan	ALLOWED	ALLOWED	ALLOWED	ALLOWED

6. REST API Reference & System Credentials

Base API Endpoint: http://localhost:5000/api

HTTP Method & Route	Protected / Roles	Description & Functionality
POST /api/auth/login	Public	Authenticates email/password, returns JWT token & user profile
GET /api/auth/me	Bearer Auth	Returns current authenticated user details
GET /api/customers	Auth (All Roles)	Paginated list with ?search=, ?status=, ?customerType=
POST /api/customers/:id/followups	ADMIN, SALES, ACCOUNTS	Logs CRM client interaction note and updates next date
GET /api/products/low-stock	Auth (All Roles)	Returns products where currentStock <= minimumStock

POST /api/stock-movements	ADMIN, WAREHOUSE	Records direct IN/OUT stock movement with audit reason
POST /api/challans	ADMIN, SALES	Creates DRAFT challan with snapshot items (No stock change)
POST /api/challans/:id/confirm	Auth (All Roles)	Executes atomic stock deduction transaction (\$transaction)
GET /api/dashboard/stats	Auth (All Roles)	Returns summary metrics, low stock alerts, upcoming follow-ups

Pre-Seeded System Test Credentials:

Role	Email Address	Password	Primary Testing Focus
ADMIN	admin@test.com	Admin@123	Full system management, deletion & user role view
SALES	sales@test.com	Sales@123	CRM follow-up logs, draft & confirm sales challans
WAREHOUSE	warehouse@test.com	Warehouse@123	Product catalog editing & direct IN/OUT stock entry
ACCOUNTS	accounts@test.com	Accounts@123	Financial records, customer detail & delivery note verification